

Question	Marking Guidance	Mark	Comments
06.1	Capsid and attachment protein;	1 (AO1)	

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06.2	1. (DNA) helicase and (DNA) polymerase; 2. (Helicase) breaks hydrogen bonds (to unwind DNA); 3. (Polymerase) condensation reactions to join (adjacent) nucleotides OR (Polymerase) forms phosphodiester bonds between (adjacent) nucleotides;	3 (3 x AO1)	1. Accept (DNA) ligase for either enzyme. 2. Reject 'hydrolyse hydrogen bonds' Reject mp3 if polymerase forming hydrogen bonds or joining complementary base pairs 3. Accept (DNA) ligase joins DNA fragments. Note: Incorrect/no enzymes named but both roles outlined = 1 mark
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06.3	Uncontrolled cell cycle/division/mitosis;	1 (AO2)	Reject meiosis Ignore growth
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06.4	For: - Needs to be given early (age 10-12) before exposure to HPV; - Will reduce transmission to girls (for when they are older); - Boys can be infected with HPV OR (Vaccination) prevents HPV infection; (Need boys) to ensure herd immunity OR (Need boys) to be above 50% vaccinated; - Boys can have increased risk of cancer (from HPV infection); Against: - Boys at less risk of cancer (from HPV infection) than females OR Cervical cancer only affects females; - May be side effects from vaccine;	4 max (4 x AO3)	Max 3 for mark points 1-5 - Accept 'Needs to be given before sexual activity' - Accept 'spread' for 'transmission'